Department of Electronic Engineering
Royal Holloway, University of London

EE1000: Creative Engineering Team Project 1

Lab 1: Practical safe soldering

Learning Outcomes:

1. To learn how to solder safely and reliably and to recognise good/bad solder joints

2. To learn how to solder and de-solder components on printed circuit boards (PCB).

1. Introduction

Soldering is a basic skill required of professional electronic engineers for 'gluing' circuit components electrically and securely together. To facilitate learning good soldering techniques, you will practice soldering components onto a circuit board and changing one of them. Finally, you will be soldering an LED flasher circuit onto a pre-designed Printed Circuit Board (PCB).

If you succeed to the satisfaction of a lab technician then you will be awarded a **soldering certificate**!

Please thoroughly read this Health and safety note BEFORE you start the lab exercise.

Health and Safety Note: **Some words of caution when soldering …**

In this lab you will be using a hot soldering iron, hot components and solder that contains flux and molten metal. Skin burns are a major risk. Be aware of what is hot so that you don't burn yourself and avoid breathing in fumes from melted solder by using the extractor fan provided with your work close to it.

Know where your soldering iron is at all times and treat it with respect - it reaches very high temperatures and can easily melt plastic and burn through clothing and fingers.

Metal conducts heat readily so avoid holding component leads and wires that you intend to solder with your fingertips – you have pliers in your toolkit. Allow time for components to cool after soldering them (blowing over them gently can facilitate cooling). Melted solder is a liquid metal - be careful not to allow it to splatter or splash around. Any molten solder that lands on your skin will burn you instantly. The wet sponge is there for wiping excess solder off the soldering iron tip and for cleaning it.

YOU MUST WEAR SAFETY GLASSES WHEN SOLDERING TO PROTECT YOUR EYES.

2. Preparing the Soldering Iron

a. Plug in and turn on the iron and wait for it to heat up (If it is temperature controlled, set it to 400 ºC for the solder you will use).

b. Apply a small amount of solder to the tip (or "bit") in order to "tin" the iron.

c. Wipe the bit gently onto the damp sponge or wire wool provided to remove any excess solder - you should now have a shiny solder bit ready for action. (Never use abrasive materials to clean the bit as this will damage the surface and cause it to wear out prematurely.)

3. Basic soldering steps (additional advice on soldering in references 1 and 2 below)

a. Apply the soldering iron tip to the parts to be soldered for a couple of seconds for them to heat up sufficiently to melt solder.

b. Touch the tip of the solder to the parts, but **not to** the tip of the iron and the solder should flow freely.

c. Once the surface of the pad is completely coated, a shiny meniscus of solder should form around the leg of the component then remove the solder and remove the soldering iron (in that order).

d. Don't move the joint for a few seconds to allow the solder to solidify (blowing gently will speed cooling). If the joint moves before it has cooled, the solder will not form a strong connection and a "dry joint" (see figure 1) can be formed. Dry joints can be fixed by re-soldering, adding a little more solder.

Good and poor solder connections to a printed circuit board

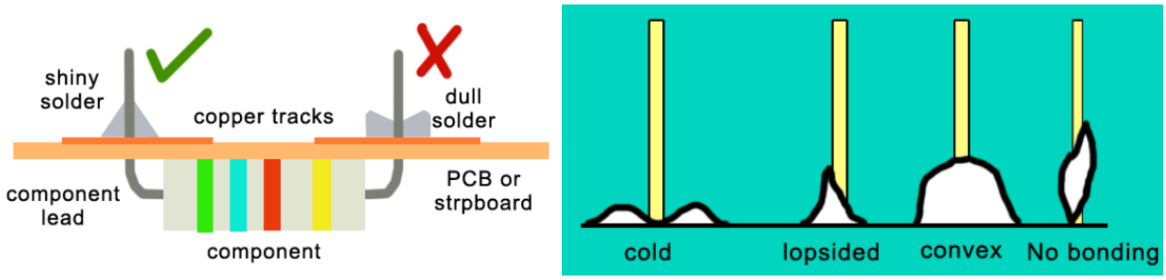

Figure 1. Examples of good and poor soldering technique (left) and typical bad or 'dry' joints when the component lead was not hot enough (right).
[*https://www.wellpcb.com/special/cold-solder-joint.html*]

A good joint has the 'conical shape' to the solder showing it has bonded along the component's lead. Notice that all the poor joints have not bonded well to the lead apart from the 'convex' result which indicates that the joint was not hot enough before it started to cool.

Try to make all your solder joints match the 'conical shape' joint in the figure.

4. Soldering components onto a printed circuit board (PCB) and removing them

1. Get a piece of Vero board (also called strip board), a 1 kΩ resistor, a red LED and cut yourself two pieces of single strand wire (about 15 cm in length - green for 'ground' and red for '+5v'). Connect the circuit in figure 2 taking care to connect the red LED the right way round (notice the lengths of its connecting wires in figure 2).

Test the circuit using the bench power supply – set its voltage to 5V and its current (limit) to 0.1A. Connect your circuit to the CH1 output and turn its "Output" to "on".

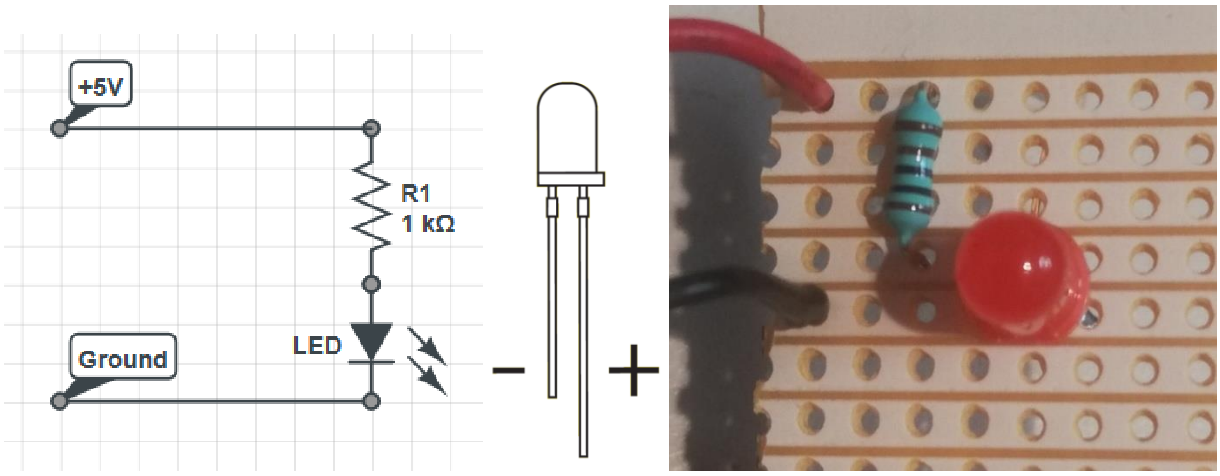

Figure 2: A simple LED circuit (left), diagram of an LED to enable its polarity to be identified - the positive lead or **anode** has the longest lead and the negative lead or **cathode** has a flat edge on the casing of the LED and components on stripboard (right).

Oh sorry, I did not mean you to use a red LED in this circuit. Use a *solder sucker* to remove the red LED and replace it with an LED of a different colour. Ask if you need help using a solder sucker. Test the circuit.

5. Making up a pre-designed circuit

You will build an astable oscillator with a buffer circuit with two LEDs which comes in kit form. Figure 3 shows the schematic of the circuit and figure 4 shows the printed circuit board (PCB) before and after the components have been added.

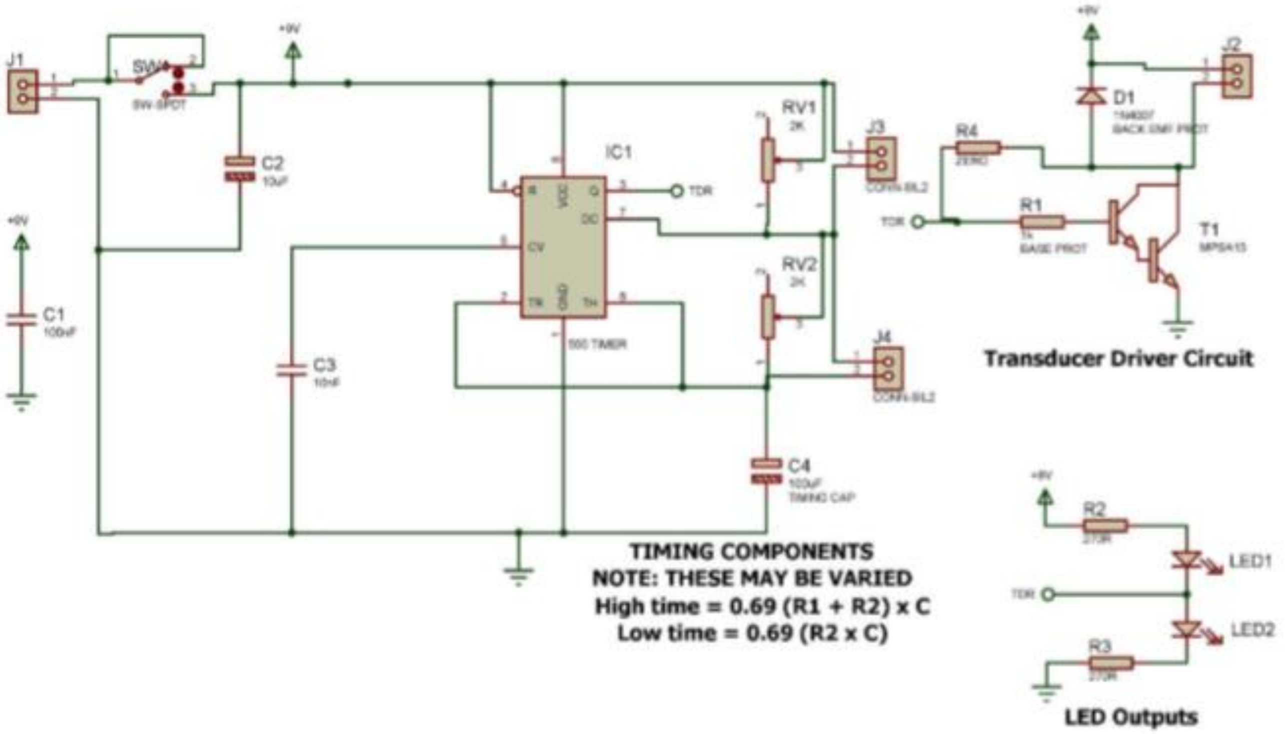

Figure 3. Circuit diagram of the astable oscillator [More details in reference 3 below]

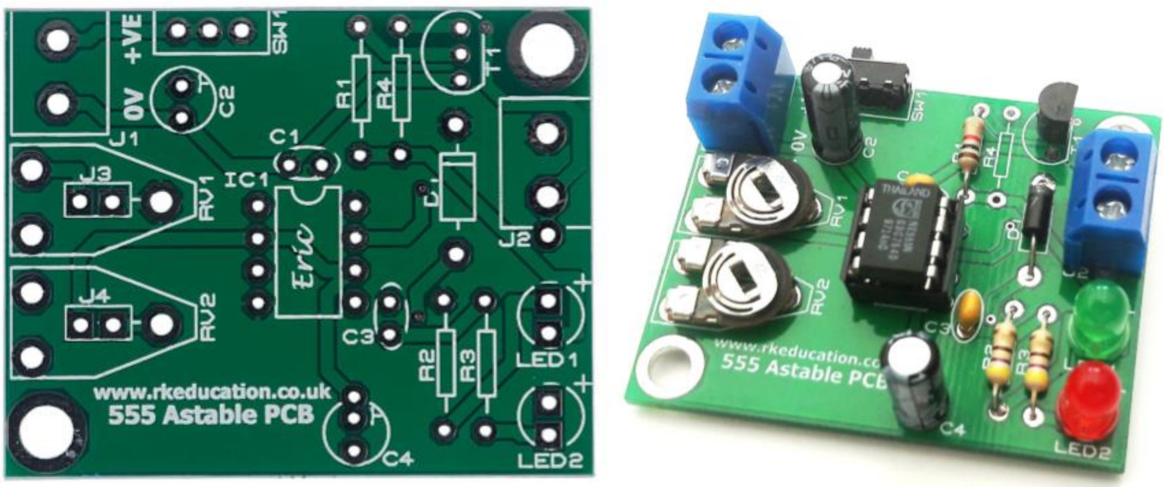

Figure 4. The unpopulated (left) and populated (right) PCB of the 555 Astable oscillator with silk screen. [Rapidonline17]

List of components:-

• T1 – 2N3904 Transistor* • C1, C2 & C3 – not used • C4 – 33µF electrolytic capacitor* • R1 – 1kΩ resistor (brown, black, red, gold) • R2, R3 – 470Ω resistor (yellow, violet, brown, gold) • R4 – not used • D1 – 1N4007 diode* • IC1 – IC holder for 555 timer IC* • RV1, RV2 – 2kΩ variable resistor • SW1 – power switch • LED1 – Red LED* • LED2 – Green LED* • Output – 2-pin Terminal Block

* These components are polarised and must be inserted the correct way around (see figures 2 and 4).

Procedure for Construction

When constructing a circuit, solder the least thermally sensitive components first as they will get heated by the soldering iron more often (e.g. resistors, switches and capacitors). Components such as ICs, transistors and diodes should be soldered last as they are most susceptible to thermal damage (if you are using an IC holder then this can be soldered at any time). It is therefore suggested that you solder the components in the following order:-

1. R1, R2 and R3 taking care with resistor colour codes
2. IC1 holder only (do not add 555 IC yet)
3. SW1 and battery clip
4. RV1 and RV2 5. LED1 and LED2 (noting the long lead is positive)
5. T1 and D1 7. C4 (take care with polarity, the negative lead is shortest)
6. Output terminal block
7. Finally, after checking the soldering over, add the 555 IC to its holder checking it's the right way up!

Testing the Circuit

Attach a PP3 9V battery to the battery clip and switch SW1 to on – hopefully if all is well the LEDs will flash. If not, then switch off and unplug the battery and check the circuit for dry joints and crossed tracks, also check polarity of the LEDs and the IC chip is correctly oriented. Ask a member of staff to assist you if you get stuck.

When your circuit is working, you can adjust the frequency of the oscillator by varying the settings of RV1 and RV2, they also affect the length of time for which each LED is on (called the "duty-cycle").

When you are happy that your circuit works correctly, you may show it to the lab technician who will assess your board and issue you with a soldering certificate if s/he is satisfied with your work!

References

1. [Instructables17] http://www.instructables.com/id/How-To-Solder-17/ (accessed 22/09/21)
2. [Evilmadscientist17] http://www.evilmadscientist.com/2015/how-not-to-solder/ (accessed 22/09/21)
3. [Rapidonline17] https://www.rapidonline.com/rk-education-555-timer-astableproject-with-drive-circuit-70-6018 (accessed 22/09/21)